



CS 2001: Data Structure

Assignment No. 4

Instructions:

1. Submit your assignment within the due date and time. Copy all your code to an MS Word file along with screenshots of your output and submit it. Late submission will result in a deduction of marks. No excuse or resubmission is permissible.
2. Mention your names and roll numbers clearly on your document.
3. Name your zip or folder/file, that you want to submit, according to the following format: **DS_A4_RollNo_FirstName**
4. Solve each task of the assignment **individually** and on your own.
5. **Plagiarized code will be penalized by a negative marking.**

Question No. 1: Graphs (Robot Navigation System)

Design and implement a robot navigation system where the starting state and the goal state have been given which are basically the coordinates of the grid of size 15x15 as shown in Figure 1. For example, the start state has the coordinates (1,2) and the goal state has the coordinates (15,14).

Consider the **following assumptions** during the implementation of the robot navigation system:

- The robot can only **move**,
 - **Up** one cell with step cost 2,
 - **Right** one cell with cost 2,
 - **Diagonally Up** towards the right with cost 3.
- The robot cannot move downward one cell.
- The obstacles are color-coded and the robot cannot be in those cells.
- The system will take a text file as input having information for the provided grid.

Your task is to **implement** the following algorithms,

1. Breadth-first search
2. Depth-first search
3. Dijkstra algorithm to find the shortest path between the start and the goal states.

Final Outputs:

Your designed system should output the followings:

- (a) Suppose that collisions are resolved by using linear probing. The integer key values listed below are to be inserted, in the order given. Show the initial slot (the slot to which the key hashes, before any probing), the probe sequence (if any) for each key, and the final contents of the hash table after the following key values have been inserted in the given order:

Key Value	Initial Slot	Probe Sequence
43		
23		
1		
0		
15		
31		
4		
7		
11		
3		

Final Hash Table:

Slot	0	1	2	3	4	5	6	7	8	9	10

Question No. 3: Closed Hashing

Consider the following keys:

17, 26, 15, 9, 11, 43, 75, 19, 35, 45, 55, 9, 10, 21, 61, 23

Implement the hash table with hash table size 15. However, in this question you will consider the following collision resolution techniques:

- Linear Probing
- Quadratic Probing
- Double Hashing

Additionally, also consider the rehashing in all parts considering the load factor of 70%, you may need to keep track of the node count for this.

Question No. 4: Open Hashing

In this question, consider the same keys as in question 1. Implement the hash table with closed hashing using each of the following collision resolution techniques:

- (a) Chaining
- (b) Bucketing

However, for bucketing, you can consider three buckets for your hash table. Additionally, you can also use rehashing if you feel any need for it. The criteria for rehashing will be same as in